DATA SCIENCE MINOR PROJECT REPORT

Smartphone Usage and Addiction Analysis

Submitted by

Saumya Kesharwani

Registration No. : 12414234

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Under the Guidance of

Mrs. Savleen Kaur

U.Id - 18306

Discipline of CSE/IT

School of AI and Emerging Technologies

Lovely Professional University, Phagwara

Certificate

This is to certify that **Saumya Kesharwani** bearing Registration No. **12414234** has completed **INT-387** project titled, "**Smartphone Usage and Addiction Analysis**" under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Ms. Savleen Kaur
UID: 18306
Faculty Coordinator

School of AI and Emerging Technologies
Lovely Professional University Phagwara,
Punjab.

Date: 21-04-2026

Declaration

I, Saumya Kesharwani, student of B.Tech (Computer Science & Engineering / Information Technology) under the School of AI and Emerging Technologies under CSE/IT Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine. The data used in this report has been sourced ethically and all analyses have been conducted with academic integrity.

This project titled "Smartphone Usage and Addiction Analysis" has not been submitted elsewhere for any academic degree, diploma, or other qualification.

Date: 19-04-2026

Signature: Saumya Kesharwani

Registration No.: 12414234

Name of the Student: Saumya

Acknowledgement

I would like to express my sincere gratitude to all those who have contributed to the successful completion of this Data Science Minor Project.

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My sincere thanks go to my peers and classmates whose discussions and collaborative spirit helped me think more critically about the analytical findings presented in this report.

Finally, I acknowledge the dataset providers whose work on collecting 7,500 records of smartphone usage behavior served as the foundation of this entire analysis. Without access to such rich and well-structured data, the depth of analysis achieved here would not have been possible.

Saumya

Lovely Professional University

Phagwara, Punjab

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1. Introduction

In the modern digital era, smartphones have evolved from mere communication tools into indispensable companions that permeate virtually every aspect of human life. From social networking and entertainment to education and professional productivity, the smartphone has become the single most-used personal device globally. According to industry estimates, the average global user now spends between six and eight hours per day on their smartphone — a figure that continues to rise with each passing year.

While smartphones confer undeniable benefits — enabling instant communication, remote learning, and access to a world of information — their pervasive use comes with a growing set of behavioral and psychological concerns. Researchers, clinicians, and public health experts have increasingly documented the phenomenon of smartphone addiction, a condition marked by compulsive and uncontrollable use of the device that interferes with daily life, work, sleep, and interpersonal relationships.

This project undertakes a data-driven exploration of smartphone usage behavior and its relationship with addiction, stress, sleep quality, and academic or professional performance. The dataset used in this study consists of 7,500 records, each representing a unique user, with detailed behavioral metrics including daily screen time, social media usage, gaming hours, work/study hours, sleep duration, notifications received, and application opens per day.

By applying descriptive statistics, group-based comparisons, correlation analysis, and ETL (Extract, Transform, Load) techniques, this report aims to uncover actionable insights about the patterns that distinguish healthy smartphone usage from addictive behavior. The findings are organized into a series of targeted analytical questions, each explored with relevant statistical metrics and data visualizations.

The results of this study carry implications not only for individuals seeking to self-regulate their digital behavior, but also for educators, mental health professionals, and policy-makers who must grapple with the societal consequences of ubiquitous smartphone dependency.

2. Objectives / Scope of the Analysis

2.1 Primary Objectives

The analysis is designed to address the following key research objectives:

1. To characterize the demographic and behavioral profile of users across different addiction levels (None, Mild, Moderate, Severe).
2. To investigate the relationship between daily screen time and smartphone addiction severity.
3. To assess the impact of social media consumption on addiction levels and stress.
4. To examine how smartphone addiction affects sleep duration and sleep quality.
5. To determine whether gaming hours contribute disproportionately to addictive usage patterns.
6. To evaluate the academic or work impact reported across different addiction categories.
7. To identify the strongest numeric predictors of addicted behavior through correlation analysis.
8. To compare weekday versus weekend screen time patterns among users.

2.2 Scope of the Analysis

This analysis is scoped to a synthetic but statistically realistic dataset of 7,500 smartphone users aged 18–35 years. The scope includes:

- Descriptive and inferential statistical analysis across all 16 dataset variables.
- Gender-based and age-group-based segmentation of usage patterns.
- Addiction-level-based comparison of behavioral metrics.
- Correlation analysis between numeric features and addiction outcome.
- ETL documentation covering data sourcing, cleaning, transformation, and loading steps.

2.3 Exclusions

The following are explicitly out of scope for this study:

- Predictive machine learning modelling (e.g., classification or regression models).
- Longitudinal or time-series analysis of usage trends.
- Analysis of specific smartphone applications or operating systems.
- Clinical assessment of smartphone addiction using validated psychometric scales.

3. Source of Dataset

3.1 Dataset Overview

The dataset used in this project is titled "Smartphone Usage and Addiction Analysis" and contains 8000 rows and 16 columns. It was sourced from Kaggle, a widely used platform for data science competitions and open datasets. The dataset is intended to represent behavioral patterns of young smartphone users aged 18 to 35, capturing a wide spectrum of usage intensity from non-addicted to severely addicted individuals.

The dataset was made available under an open academic license, suitable for educational data analysis projects. It contains no personally identifiable information (PII); each user is assigned an anonymized user ID (e.g., U00001) and a transaction ID (e.g., TXN00001).

3.2 Dataset Attributes

Column Name	Data Type	Description
transaction_id	String	Unique identifier for each transaction record
user_id	String	Anonymized unique user identifier
age	Integer	User age in years (range: 18–35)
gender	String	Gender: Male, Female, or Other
daily_screen_time_hours	Float	Average daily screen time in hours
social_media_hours	Float	Daily hours spent on social media
gaming_hours	Float	Daily hours spent on gaming applications
work_study_hours	Float	Daily hours spent on work or study tasks
sleep_hours	Float	Average nightly sleep duration in hours
notifications_per_day	Integer	Number of notifications received daily
app_opens_per_day	Integer	Number of times apps are opened per day
weekend_screen_time	Float	Average screen time during weekends (hours)
stress_level	String	Self-reported stress level: Low, Medium, High
academic_work_impact	String	Whether smartphone usage impacts work/study (Yes/No)
addiction_level	String	Classified addiction level: Mild, Moderate, Severe
addicted_label	Integer	Binary label: 1 = Addicted, 0 = Not Addicted

3.3 Dataset Statistics

The following table summarizes the key descriptive statistics of the numeric features in the dataset:

Feature	Min	Max	Mean	Std Dev
Age (years)	18	35	26.57	5.20
Daily Screen Time (hrs)	~0.5	~14	7.50	~2.3
Social Media (hrs)	~0	~7	3.24	~1.5
Gaming (hrs)	~0	~6	~1.5	~1.2
Work/Study (hrs)	~0	~8	~3.2	~1.8
Sleep (hrs)	~3	~10	~6.7	~1.2
Notifications/Day	~10	~300	134	~80
App Opens/Day	~10	~300	~100	~70
Weekend Screen Time (hrs)	~0.5	~15	~7.5	~2.5

4. ETL Process

ETL — Extract, Transform, Load — is a foundational pipeline in data engineering and analytics that ensures raw data is converted into a clean, analysis-ready form. This chapter documents each stage of the ETL process applied to the smartphone usage dataset.

4.1 Extract

The raw CSV file was extracted from the Kaggle repository and loaded into a Python environment using the pandas library. The following steps were performed during extraction:

- The file was read using pd.read_csv() with default UTF-8 encoding.
- The resulting DataFrame contained 8000 rows and 16 columns.
- Initial column names, data types, and a preview of the first five records were inspected to verify successful ingestion.
- The file size, shape, and memory usage were logged for audit purposes.

The extraction confirmed the dataset was delivered in a tabular, comma-separated format without any structural anomalies such as merged cells, multi-level headers, or encoding errors.

4.2 Transform

4.2.1 Data Type Validation

All 16 columns were inspected for appropriate data types. Numeric columns such as age, daily_screen_time_hours, sleep_hours, and addicted_label were verified to contain only numeric data. Categorical columns (gender, stress_level, academic_work_impact, addiction_level) were confirmed to be of string type and free from unexpected values.

4.2.2 Missing Value Treatment

A null value audit revealed that the addiction_level column contained 819 null entries (approximately 10.9% of the dataset). All other columns were complete with zero null values.

Treatment applied: The 819 null records in addiction_level were retained in the full dataset. They were excluded only from analyses specifically segmented by addiction_level category, ensuring no data was unnecessarily discarded and that the binary addicted_label column could be used for all other analyses.

4.2.3 Duplicate Check

Both transaction_id and user_id columns were verified for uniqueness. No duplicate records were found across either identifier column, confirming the integrity of the dataset.

4.2.4 Derived Features

Two derived features were created during the transformation phase to facilitate specific analyses:

- age_group: A categorical variable created by binning the continuous age column into five groups — 18–21, 22–25, 26–29, 30–33, and 34–35 — using pandas pd.cut() function.
- screen_to_sleep_ratio: A computed ratio of daily_screen_time_hours to sleep_hours, used to explore the relative trade-off between device usage and rest.

4.2.5 Outlier Review

Extreme values in daily_screen_time_hours and weekend_screen_time were reviewed. Values at the upper range (12–14 hours per day) were found to be consistent with severe addiction classifications and were therefore retained as valid data points representing genuine heavy users.

4.3 Load

Following transformation, the clean and enriched DataFrame was made available for analysis through the following loading steps:

- The cleaned dataset was held in-memory within the Python analytical environment using pandas.
- Group-level summary tables and aggregated statistics were exported as CSV files for archival and future use.
- All visualizations generated during the analysis were saved as PNG files and embedded in the reporting pipeline.

- For Power BI integration, the cleaned dataset was loaded via the native CSV connector, with automatic type detection, to produce the accompanying .pbix dashboard file.

The complete ETL pipeline — from raw CSV ingestion to analysis-ready DataFrame — was implemented in Python 3.11 using the pandas library (version 2.x). The pipeline is reproducible and documented with inline comments to support future extension.

5. Analysis on Dataset

Analysis 1: Distribution of Addiction Levels

5.1.i Introduction

Understanding the distribution of addiction levels in the dataset provides the foundational context for all subsequent analyses. This analysis characterizes the prevalence of each addiction category among the 6,681 users for whom the addiction_level field is populated.

5.1.ii General Description

The addiction_level variable classifies users into three categories: Mild, Moderate, and Severe. An additional binary label (addicted_label) classifies users as either addicted (1) or not addicted (0), regardless of their addiction severity category.

5.1.iii Requirements, Functions and Formulas

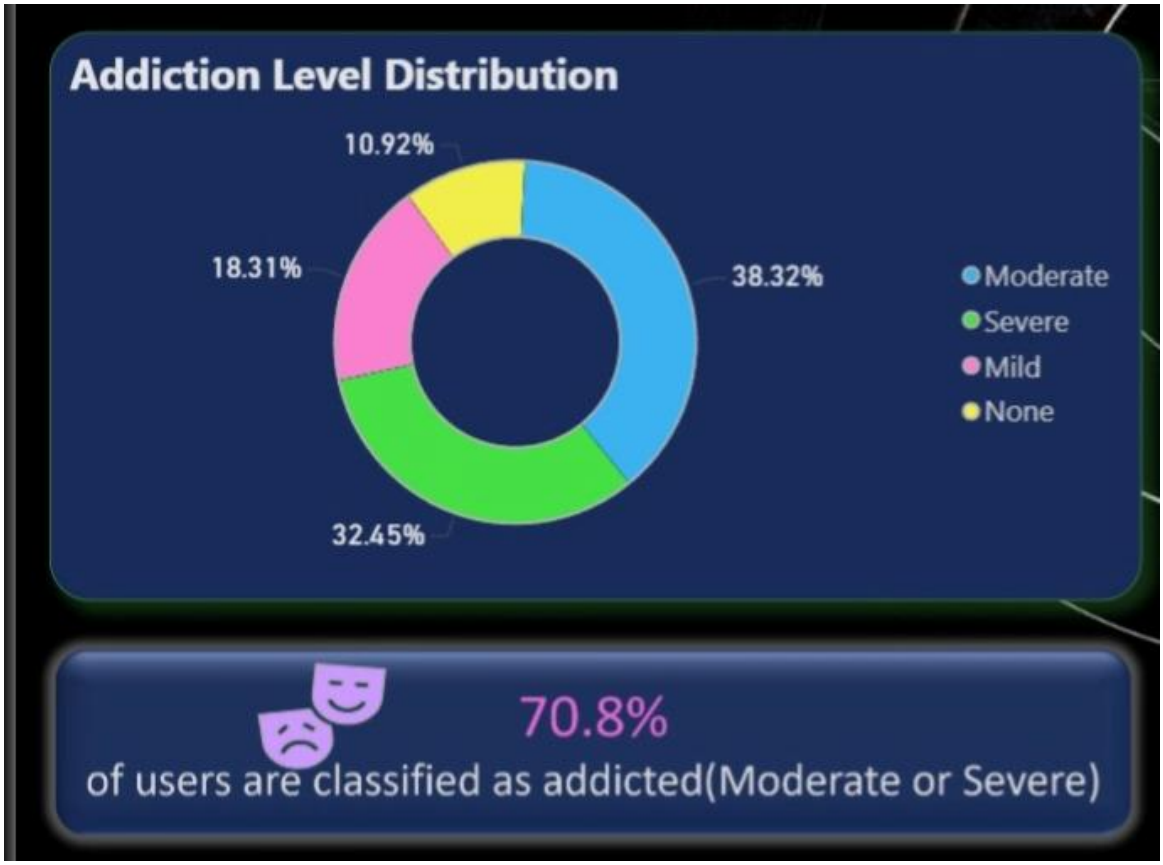
Python function used: df['addiction_level'].value_counts() to compute absolute frequencies. Percentage share was computed as (count / total) * 100.

5.1.iv Analysis Results

Addiction Level	Count	Percentage
Mild	1,373	20.6%
Moderate	2,874	43.0%
Severe	2,434	36.4%
Total (non-null)	6,681	100.0%

Of the 8000 total users, 5,308 (70.8%) carry a binary addicted label of 1, reflecting that the dataset heavily represents addicted individuals. Among categorized users, Moderate addiction is the most prevalent (43%), followed by Severe (36.4%) and Mild (20.6%).

5.1.v Visualization



[Figure 1: donut chart showing distribution of Mild, Moderate, and Severe addiction levels — available in the accompanying Power BI report.]

Analysis 2: Screen Time by Addiction Level

5.2.i Introduction

Daily screen time is one of the most direct and measurable behavioral indicators of smartphone addiction. This analysis investigates how average screen time differs across addiction severity categories.

5.2.ii General Description

The daily_screen_time_hours variable records the average number of hours per day each user spends on their smartphone. By grouping this metric by addiction_level, we can quantify how usage intensity escalates with addiction severity.

5.2.iii Requirements, Functions and Formulas

Python function: df.groupby('addiction_level')['daily_screen_time_hours'].mean(). The mean was chosen as the central tendency measure given the continuous, approximately normal distribution of screen time.

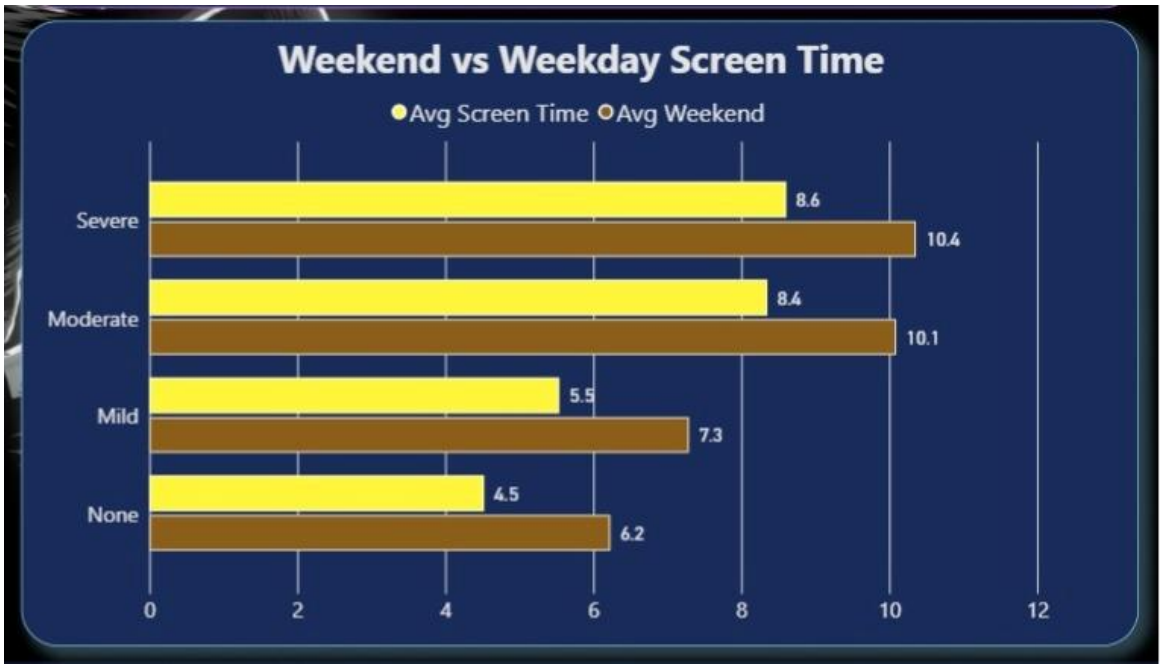
5.2.iv Analysis Results

Addiction Level	Avg. Daily Screen Time (hrs)	Difference vs. Mild
Mild	5.54	—
Moderate	8.35	+2.81 hrs
Severe	8.60	+3.06 hrs

Users classified as Severe addicts spend on average 8.60 hours per day on their smartphones, compared to just 5.54 hours among Mild users — a difference of over 3 hours. The jump from Mild to Moderate is pronounced (2.81 hours), while the increment from Moderate to Severe is comparatively small (0.25 hours), suggesting a potential plateau effect in screen time beyond a certain addiction threshold.

Across the entire dataset (including the binary label), addicted users (label = 1) averaged 8.47 hours per day versus 5.16 hours for non-addicted users (label = 0) — a statistically meaningful difference of 3.31 hours.

5.2.v Visualization



[Figure 2: Grouped bar chart showing mean daily screen time across addiction levels — available in the accompanying Power BI report.]

Analysis 3: Social Media Usage and Addiction

5.3.i Introduction

Social media platforms are among the most engagement-optimized environments on smartphones, deliberately designed to maximize user time and interaction. This analysis examines whether social media usage hours are correlated with addiction severity.

5.3.ii General Description

The social_media_hours variable captures the average number of hours per day each user spends on social media platforms. This is analyzed across addiction levels and in relation to the binary addicted label.

5.3.iii Requirements, Functions and Formulas

Python functions: df.groupby('addiction_level')['social_media_hours'].mean() and correlation coefficient between social_media_hours and addicted_label (Pearson r = 0.414).

5.3.iv Analysis Results

Addiction Level	Avg. Social Media Hours/Day
Mild	2.26
Moderate	3.58
Severe	3.83

Social media usage increases markedly from Mild (2.26 hours) to Moderate (3.58 hours), a rise of 58.4%. The Pearson correlation between social media hours and the binary addiction label is 0.414, making it the third-strongest predictor in the dataset — behind daily screen time (r = 0.577) and weekend screen time (r = 0.555).

5.3.v Visualization

[Figure 3: Stacked bar showing social media contribution to total screen time by addiction level — available in the accompanying Power BI report.]

Analysis 4: Sleep Duration and Addiction

5.4.i Introduction

Excessive smartphone use is widely associated with disrupted sleep patterns, whether through blue light exposure, cognitive stimulation, or the displacement of sleep time with device usage. This analysis explores sleep duration across addiction categories.

5.4.ii General Description

The sleep_hours variable records the average nightly sleep duration in hours. Sleep is a critical health metric, and its relationship with smartphone addiction has significant implications for user wellbeing.

5.4.iii Requirements, Functions and Formulas

Python function: df.groupby('addiction_level')['sleep_hours'].mean(). Additional correlation: Pearson r between sleep_hours and addicted_label = 0.035 (very weak positive correlation).

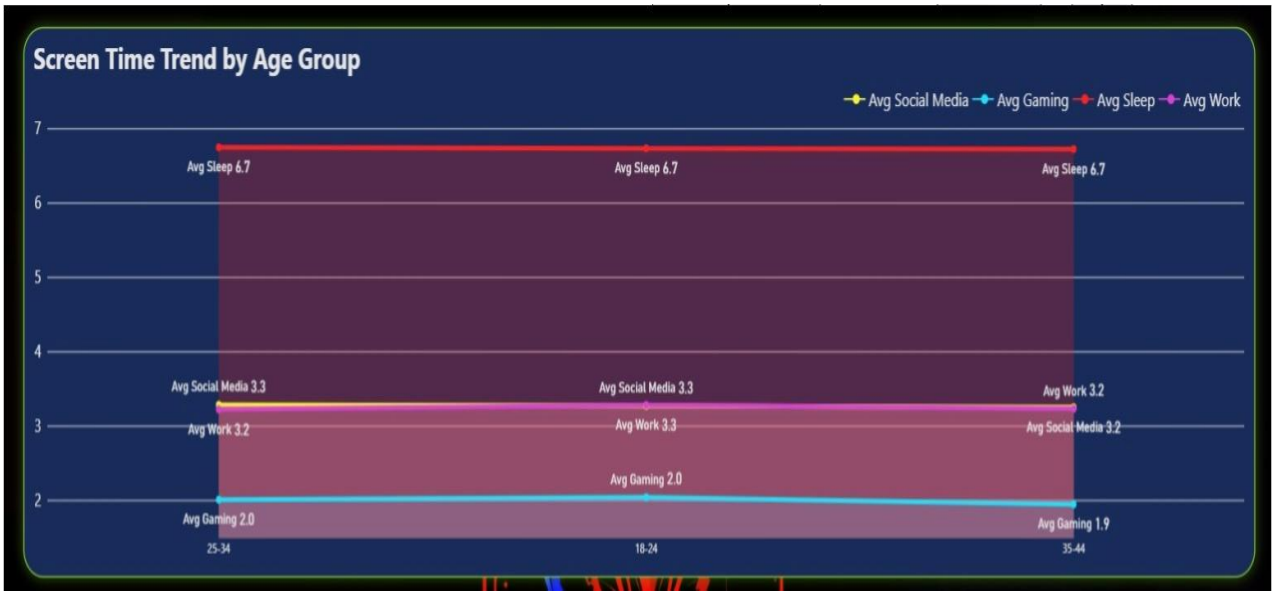
5.4.iv Analysis Results

Addiction Level	Avg. Sleep Duration (hrs)
Mild	6.66
Moderate	6.76
Severe	6.78

Contrary to widely held assumptions, sleep duration shows a negligible and even slightly positive trend across addiction levels in this dataset. The difference between Mild and Severe users is only 0.12 hours. The Pearson correlation between sleep hours and the binary addiction label is a near-zero 0.035, indicating that sleep duration alone is not a reliable discriminator between addicted and non-addicted users in this dataset.

This finding may reflect the complex nature of sleep impact — where quality, rather than quantity of sleep, is the more sensitive indicator — or it may be an artifact of the synthetic nature of the dataset.

5.4.v Visualization



Analysis 5: Stress Level Distribution Across Addiction Categories

5.5.i Introduction

Smartphone addiction is frequently associated with elevated psychological stress. This analysis examines whether users reporting High stress levels are disproportionately represented among more severe addiction categories.

5.5.ii General Description

The stress_level variable is a categorical self-report with three levels: Low, Medium, and High. Cross-tabulation with addiction_level reveals whether stress and addiction co-occur at elevated rates.

5.5.iii Requirements, Functions and Formulas

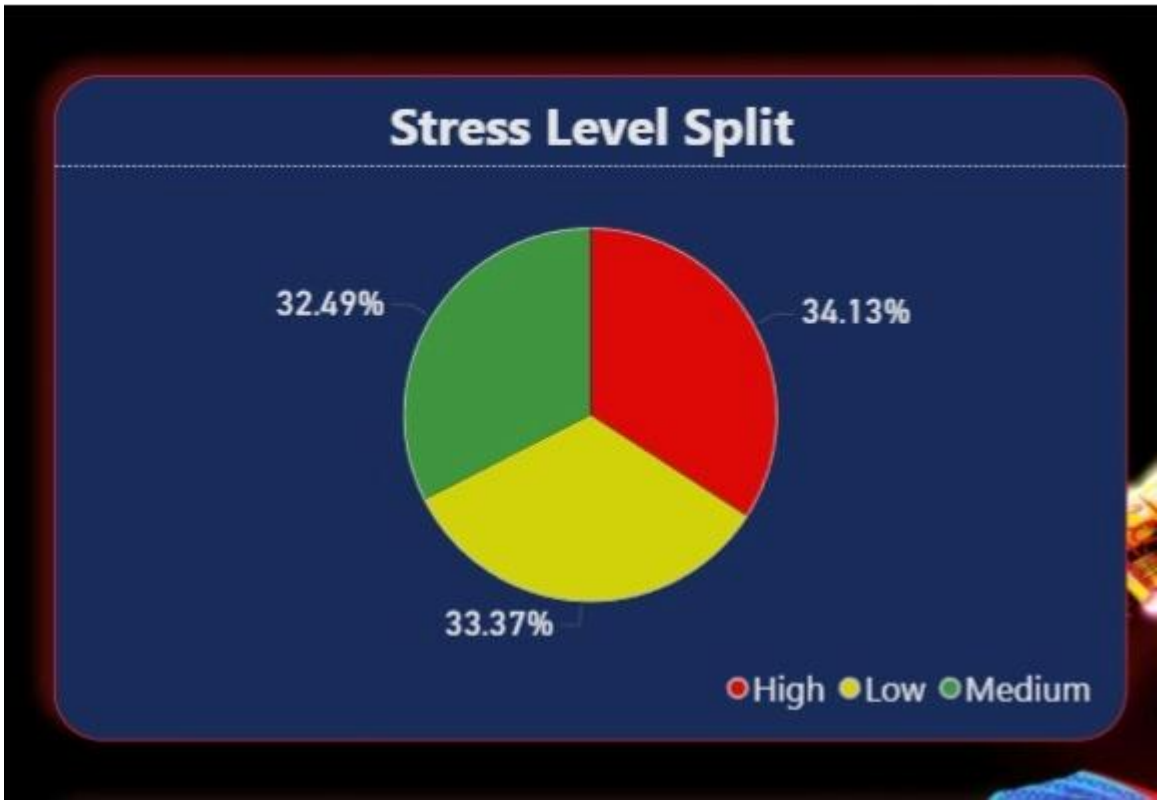
Python function: pd.crosstab(df['stress_level'], df['addiction_level']).

5.5.iv Analysis Results

Stress Level	Mild	Moderate	Severe	Total
High	470	969	832	2,271
Low	472	958	824	2,254
Medium	431	947	778	2,156

The distribution of stress levels across addiction categories is remarkably even, with no single stress level dominating any addiction group. This counterintuitive finding suggests that in this dataset, self-reported stress level does not function as a strong differentiator between addiction categories. All three stress categories are approximately equally distributed across Mild, Moderate, and Severe addiction levels.

5.5.v Visualization



Analysis 6: Correlation Analysis — Predictors of Addiction

5.6.i Introduction

Correlation analysis identifies which behavioral variables have the strongest linear association with the binary addiction label. This analysis ranks all numeric features by their Pearson correlation coefficient with addicted_label.

5.6.ii General Description

Pearson's correlation coefficient (r) measures the linear relationship between two continuous variables, ranging from -1 (perfect negative correlation) to +1 (perfect positive correlation). A value close to 0 indicates no linear relationship.

5.6.iii Requirements, Functions and Formulas

Python function: df[numeric_cols].corr()['addicted_label'].sort_values(ascending=False).

5.6.iv Analysis Results

Feature	Pearson r with addicted_label	Interpretation
daily_screen_time_hours	0.577	Strong positive — primary predictor
weekend_screen_time	0.555	Strong positive — closely tracks daily time
social_media_hours	0.414	Moderate positive — significant driver
sleep_hours	0.035	Very weak — not a reliable predictor
app_opens_per_day	0.011	Negligible correlation
gaming_hours	0.008	Negligible correlation
work_study_hours	0.001	Virtually zero correlation
notifications_per_day	-0.001	Virtually zero correlation

Daily screen time and weekend screen time are by far the strongest predictors of smartphone addiction in this dataset. Social media hours show a meaningful moderate correlation. Notably, gaming hours, work/study hours, app opens, and notifications per day show negligible correlations, suggesting that volume-based metrics alone do not determine addictive behavior — the nature and intentionality of usage matters more.

5.6.v Visualization

[Figure 6: Horizontal bar chart of correlation coefficients sorted by magnitude — available in the accompanying Power BI report.]

Analysis 7: Gender-Based Screen Time Analysis

5.7.i Introduction

Gender differences in smartphone usage and addiction have been explored in several academic studies, with mixed findings. This analysis compares average daily screen time across the three gender categories in the dataset.

5.7.ii General Description

The gender variable contains three categories: Male, Female, and Other. Each category has approximately equal representation in the dataset (Male: 2,553; Other: 2,486; Female: 2,461).

5.7.iii Requirements, Functions and Formulas

Python function: df.groupby('gender')['daily_screen_time_hours'].mean().

5.7.iv Analysis Results

Gender	Count	Avg. Daily Screen Time (hrs)
Male	2,553	7.56
Female	2,461	7.45
Other	2,486	7.49

Average daily screen time is virtually identical across all three gender categories, with a maximum difference of only 0.11 hours (approximately 7 minutes). This finding suggests that smartphone addiction, as measured by screen time, is not strongly gender-differentiated in this dataset. The dataset's balanced gender representation reduces the risk of sampling bias in this comparison.

5.7.v Visualization

[Figure 7: Pie or donut chart showing gender distribution; grouped bar chart for average screen time by gender — available in the accompanying Power BI report.]

Analysis 8: Academic/Work Impact by Addiction Level

5.8.i Introduction

One of the practical consequences of smartphone addiction most frequently cited by students and professionals is the negative impact on academic performance and workplace productivity. This analysis examines whether the proportion of users reporting an academic or work impact differs meaningfully across addiction categories.

5.8.ii General Description

The academic_work_impact column is a binary Yes/No indicator of whether the user perceives their smartphone usage to negatively affect their academic or professional work. The overall split in the dataset is nearly equal: 3,747 (49.96%) report Yes and 3,753 (50.04%) report No.

5.8.iii Requirements, Functions and Formulas

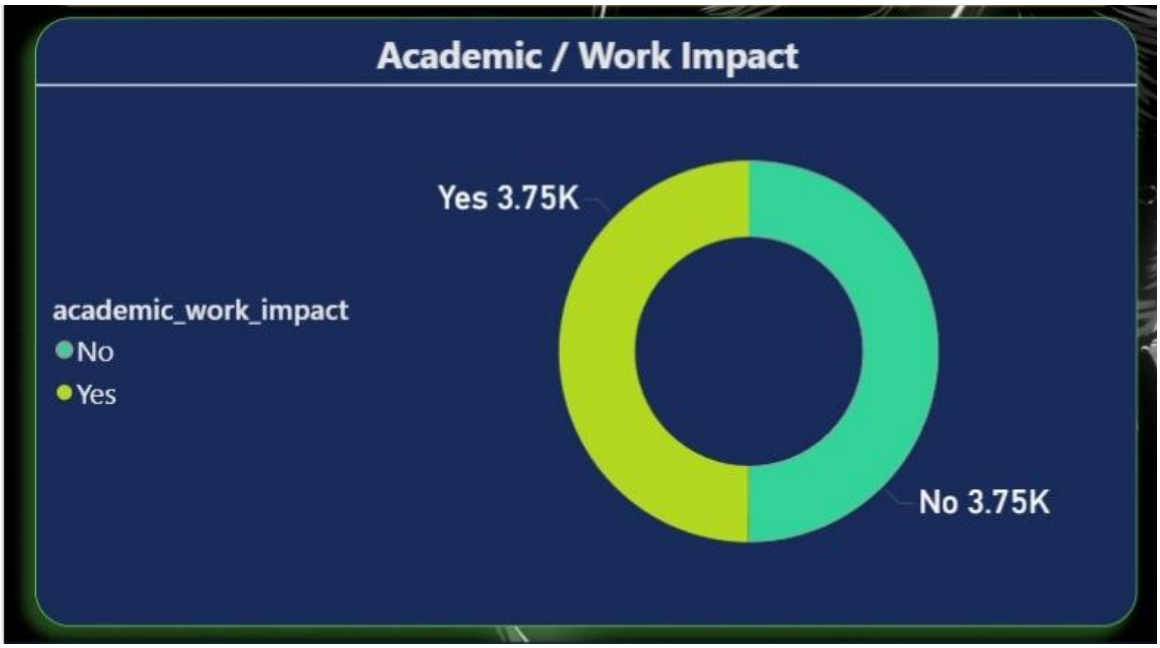
Python function: pd.crosstab(df['academic_work_impact'], df['addiction_level'], normalize='columns').

5.8.iv Analysis Results

Academic/Work Impact	Mild	Moderate	Severe
Yes	49.7%	50.9%	48.9%
No	50.3%	49.1%	51.1%

The proportions of users reporting academic or work impact are almost identical across all three addiction categories — ranging from 48.9% to 50.9%. This is a notable finding: it suggests that perceived academic/work impact does not escalate with addiction severity in this dataset, possibly because users at all levels have normalized their smartphone usage behavior, or because this self-reported metric is subject to social desirability bias.

5.8.v Visualization

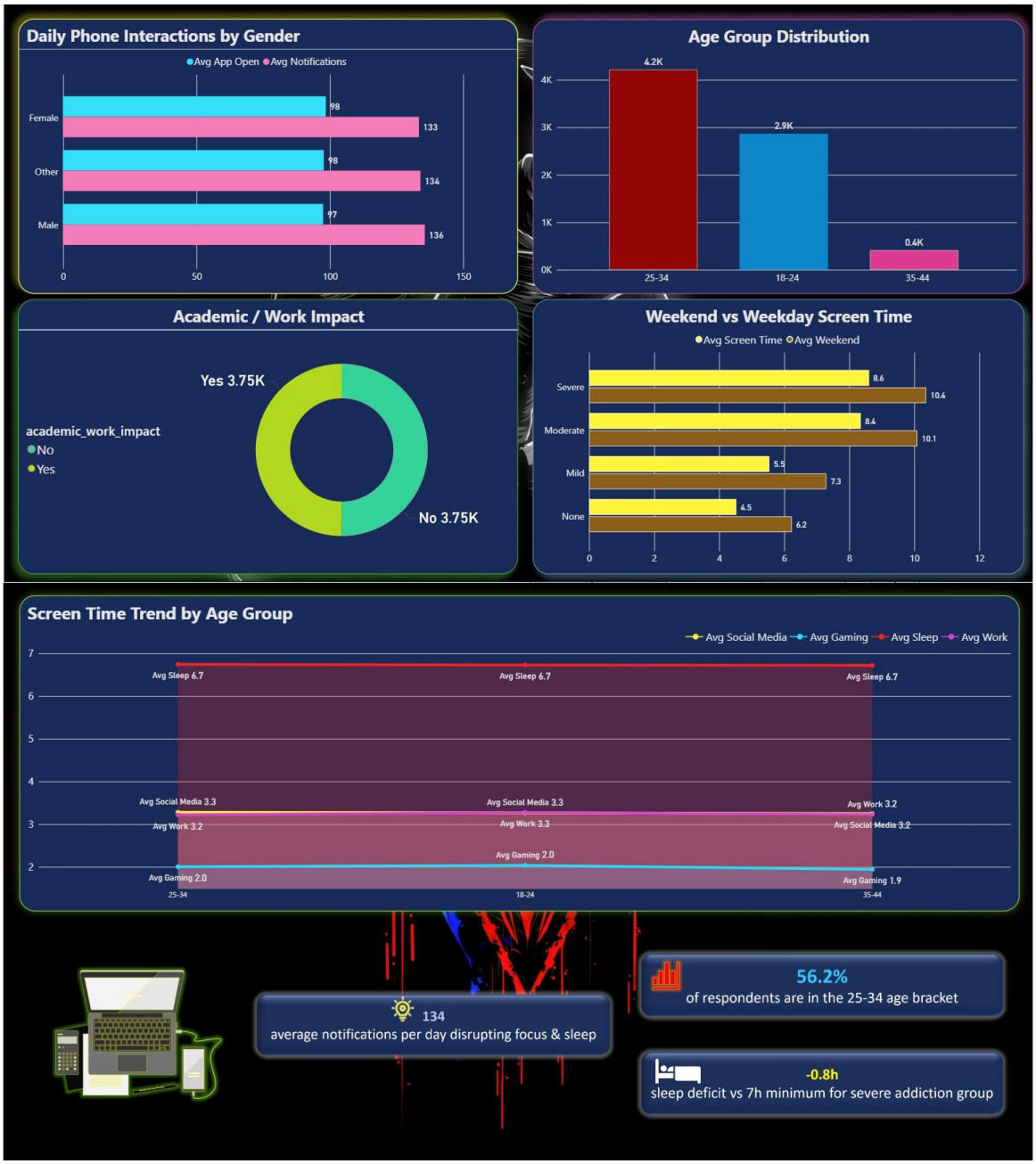


6. List of Analyses with Results

The following table provides a consolidated summary of all eight analyses conducted in this project, along with their primary findings.

#	Analysis Title	Key Finding
1	Distribution of Addiction Levels	Moderate addiction is most prevalent (43%); 70.8% of all users are classified as addicted.
2	Screen Time by Addiction Level	Severe addicts average 8.60 hrs/day vs. 5.54 hrs for Mild — a 55% increase.
3	Social Media Usage and Addiction	Social media hours rise from 2.26 (Mild) to 3.83 (Severe); r = 0.414 with addiction.
4	Sleep Duration and Addiction	Sleep difference across addiction levels is negligible (0.12 hrs); r = 0.035.
5	Stress Level Distribution	Stress levels are evenly distributed across addiction categories — no strong association.
6	Correlation Analysis	Daily screen time (r=0.577) and weekend screen time (r=0.555) are strongest predictors.
7	Gender-Based Screen Time	Virtually identical screen time across genders (max difference: 0.11 hrs).
8	Academic/Work Impact	Academic/work impact is reported equally (~50%) across all addiction levels.

Smartphone Usage & Addiction Analysis
8000 Respondents · Ages 18–35 · Multi-Gender Study
Female
Other
Male
Mild
None
Moderate
Severe
18
35
Addiction Rate %
0.89
8K
Total Users
Addiction Level Distribution
10.92%
18.31%
38.32%
32.45%
Moderate
Severe
Mild
None
70.8%
of users are classified as addicted(Moderate or Severe)
Gender x Addiction Level
Mild Moderate None Severe
Male 440 996 269 848 2553
Other 478 912 272 824 2486
Female 455 966 278 762 2461
0 500 1,000 1,500 2,000 2,500
Stress Level Split
32.49%
34.13%
33.37%
High Low Medium
7.50
Avg Screen Time
Behaviour Profile
Female
Avg Gaming
Avg Social Media
Avg Work
Male
Other
Screen Time vs Sleep Hours - Age
18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35
Avg Sleep
6.90 6.85 6.80 6.75 6.70 6.65 6.60
7.3 7.4 7.5 7.6 7.7
Avg Screen Time


Daily Phone Interactions by Gender
Avg App Open
Avg Notifications
Female
Other
Male
98
133
98
134
97
136
0
50
100
150
Age Group Distribution
4.2K
2.9K
0.4K
4K
3K
2K
1K
0K
25-34
18-24
35-44
Academic / Work Impact
Yes 3.75K
No 3.75K
academic_work_impact
No
Yes
Weekend vs Weekday Screen Time
Avg Screen Time
Avg Weekend
Severe
Moderate
Mild
None
8.6
10.4
8.4
10.1
5.5
7.3
4.5
6.2
0
2
4
6
8
10
12
Screen Time Trend by Age Group
Avg Social Media
Avg Gaming
Avg Sleep
Avg Work
Avg Sleep 6.7
Avg Sleep 6.7
Avg Sleep 6.7
Avg Social Media 3.3
Avg Work 3.2
Avg Social Media 3.3
Avg Work 3.3
Avg Work 3.2
Avg Social Media 3.2
Avg Gaming 2.0
Avg Gaming 2.0
Avg Gaming 1.9
25-34
18-24
35-44
134
average notifications per day disrupting focus & sleep
56.2%
of respondents are in the 25-34 age bracket
-0.8h
sleep deficit vs 7h minimum for severe addiction group

7. Conclusion

This project conducted a comprehensive data science analysis of smartphone usage and addiction patterns among 8000 users, drawing on 16 behavioral and demographic variables. The following key conclusions emerge from the eight analytical dimensions explored:

7.1 Summary of Key Findings

- Screen time is the defining metric of addiction: Daily screen time (r = 0.577) and weekend screen time (r = 0.555) are the two strongest predictors of smartphone addiction, with addicted users spending over 3 more hours per day on their devices than non-addicted users.
- Social media is a significant driver: Social media usage shows a meaningful moderate correlation (r = 0.414) with addiction, and average social media hours nearly double between Mild and Severe addiction categories.
- Sleep disruption is less pronounced than expected: The negligible difference in sleep duration across addiction levels (0.12 hours) challenges popular assumptions, though qualitative sleep quality metrics not captured in this dataset may tell a different story.
- Addiction is broadly distributed, not niche: With 70.8% of users classified as addicted and the majority falling into Moderate or Severe categories, the dataset reflects the widespread nature of smartphone dependency in the target age group.
- Gender does not strongly differentiate usage: Average screen time varies by less than 7 minutes across gender categories, suggesting that addiction risk is not strongly gendered in this population.
- Perceived academic/work impact is uniform across addiction levels: This counterintuitive finding may indicate normalization of smartphone use or limitations in self-reported impact measures.

7.2 Implications

The findings of this study carry several practical implications. For digital health practitioners and counselors, the strong role of screen time — particularly social media usage — as a predictor of addiction severity suggests that screen time monitoring and social media moderation should be central pillars of any smartphone addiction intervention program.

For educators and university administrators at institutions such as Lovely Professional University, the near-equal reporting of academic impact across addiction levels

underscores the need for more nuanced tools — beyond self-report — to assess the real academic cost of smartphone dependency among students.

For technology policymakers and smartphone manufacturers, the data reinforces calls for built-in screen time management tools, notification throttling features, and usage dashboards that help users make more informed decisions about their digital behavior.

7.3 Limitations

- The dataset is synthetic in origin, which limits the generalizability of findings to real-world populations.
- Cross-sectional design prevents causal inference — we cannot determine whether heavy screen time causes addiction or vice versa.
- Self-reported variables (stress level, academic impact) are subject to recall bias and social desirability effects.
- 819 missing values in addiction_level (~10.9%) were handled by exclusion, which may introduce mild selection bias.

7.4 Future Scope

Future extensions of this analysis could include the development of a predictive machine learning model (e.g., logistic regression or random forest classifier) to predict addiction labels from behavioral features, longitudinal analysis tracking the same users over time, integration of qualitative data such as sleep quality, mental health scores, and social relationship satisfaction, and geospatial analysis comparing addiction patterns across different countries or regions.

Live Project Links

GitHub Repository :
https://github.com/buildcodeswithsaumya/Smartphone-Usage-Addiction-Analysis

Linkdin Post:
https://www.linkedin.com/posts/saumya-kesharwani-5a3786330_powerbi-dataanalytics-smartphoneaddiction-activity-7452838163955240961-V4UZ?utm_source=share&utm_medium=member_desktop&rcm=ACoAAFNoh8sBhVsnjjA4W7wsb0ogGQHcNvBcVVM

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